034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

FULL WRITTEN EXAMINATION - A.Y. 2023/2024

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PART 1 - Problem 2 - Stabilisation of an LTI System using the "full state feedback" - A Solution Proposal

Tag List: `[STANDARD ASSESSMENT] [PARTIAL CREDIT] [SYMBOLIC]`

Introduction

Consider a generic LTI SISO system

$$\dot{x}(t) = A\, x(t) + B\, u(t)$$

The system dynamics can be described using the state movement, depending on the eigenvalues of the matrix A (refer to the class material **Part 2, p. 33** and **Part 3, pp. 7, 12 and 23**).

The Full State Feedback

The **full state feedback** is a feedback control strategy, allowing the placement of the closed-loop poles of the system in predetermined locations in the s-plane (s is the complex variable of the Laplace transform). Placing the poles means imposing the location of the system eigenvalues, i.e., controlling the characteristics of the state movement (through the response modes). The LTI system must be controllable in order to apply such a control strategy.

Considering the properties of a SISO LTI system, the "complete state controllability," or simply "controllability," is very important. Controllability is fundamental when solving many control problems, such as stabilising an unstable LTI system through state feedback. No further details on the controllability property or methods for verifying whether a SISO LTI system is controllable will be provided.

For simplicity, let's assume that **the LTI systems** analysed in this live-script are **controllable**.

Thus, given the controllable SISO LTI system

$$\dot{x}(t) = A\, x(t) + B\, u(t) \qquad (\star)$$

applying the full state feedback means imposing the input $u(t)$ as linear combination of the state $x(t)$:

$$u(t) = -K\, x(t) \qquad (+)$$

where K is a proper row-matrix.

What happens to the system's dynamics? Let's substitute the expression $(+)$ in Eq. $(\star)$:

$$\dot{x} = \left(A - B\, K\right)\, x(t)$$

The full-state-feedback system has a different dynamics, depending on the eigenvalues of the matrix $\left(A - B\, K\right)$

Example

Consider the SISO LTI system described by the state equations

$$\dot{x}(t) = \begin{bmatrix} 0 & -2 \\ +1 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

The system is controllable. The system's eigenvalues are:

$$p_A(\lambda) = \lambda^2 + 3\,\lambda + 2 \quad \Longrightarrow \quad \begin{cases} \lambda_1 & = & -1 \\ \lambda_2 & = & -2 \end{cases}$$

Let's apply the full state feedback:

$$u(t) = \begin{bmatrix} k_1 & k_2 \end{bmatrix} x(t)\,, \qquad k_1, k_2 \in \mathbb{R}$$

Thus, the new system dynamics is described by the state equations

$$\dot{x}(t) = \begin{bmatrix} 0 & -2 \\ +1 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} k_1 & k_2 \end{bmatrix} x(t) \quad \Longrightarrow \quad \dot{x}(t) = \begin{bmatrix} 0 & -2 \\ k_1 + 1 & k_2 - 3 \end{bmatrix} x(t)$$

The new characteristic polynomial is

$$\det\left(s\,I - \begin{bmatrix} 0 & -2 \\ k_1 + 1 & k_2 - 3 \end{bmatrix} \right) = \lambda^2 + (3 - k_2)\,\lambda + 2\,k_1 + 2$$

Note: by choosing properly the values k_1 and k_2 we can impose the preferred eigenvalues. For example

$$\begin{cases} k_1 & = & \dfrac{23}{2} \\ k_2 & = & -7 \end{cases} \quad \Longrightarrow \quad \begin{cases} \hat{\lambda}_1 & = & -5 \\ \hat{\lambda}_2 & = & -5 \end{cases}$$

Assignment

Consider the SISO LTI controllable system described by the following state equations

$$\dot{x}(t) = \begin{bmatrix} 0 & 0 & +1 \\ 0 & -2 & -1 \\ +4 & 0 & +2 \end{bmatrix} x(t) + \begin{bmatrix} +1 \\ +1 \\ +2 \end{bmatrix} u(t)$$

The actual system is unstable.

In order to stabilise the system, you should apply a full state feedback

$$u(t) = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix}\,, \quad k_1, k_2, k_3 \in \mathbb{R}$$

Your code should:

- Declare the **<u>symbolic</u>** <u>real</u> variables, named **k1**, **k2** and **k3**, corresponding to the components of the state feedback control law.

- Declare the <u>**symbolic**</u> variable **x**.
- Declare the <u>**symbolic**</u> variable **matA** corresponding to the matrix A. Declare also the symbolic variable **matB** corresponding to the matrix B.
- Verify that the actual LTI system is unstable, by determining the eigevalues of **matA**. Save the result in the symbolic array **lambdaA**.
- Compute the expression of the matrix $A + B\,K$ and assign the result to the <u>**symbolic**</u> variable **matAK**.
- Determine the characteristic polynomial of the matrix **matAK** and save its expression as the <u>**symbolic variable**</u> **pAK_x**, function of the symbolic variable **x**, already defined.
- **Note**: the polynomial **pAK_x** must depend on the state feedback parameters **k1**, **k2** and **k3**.
- Determine <u>the analytic expression</u> of the desired characteristic polynomial $d(x) = (x+10)\cdot(x+20)\cdot(x+30)$ as a polynomial ordered by descending powers of the symbolic variable **x**. You may use the MATLAB commands expand( )s. Assign the result to the symbolic variable **pD_x**.
- By comparison of the coefficients of the polynomial **pAK_x** (coefficients depending on the unknowns **k1**, **k2** and **k3**) and the corresponding coefficients of the polynomial **pD_x**, you can assembly a set of three linear equations concerning the three unknowns **k1**, **k2** and **k3**. Solve this set of equations, using the MATLAB command solve( ). Save the result in the symbolic vector **kSOL**.

Solution

```
clear variables
clc

syms k1 k2 k3 real
syms x

assumptions
```

ans = $\left(k_1 \in \mathbb{R} \quad k_2 \in \mathbb{R} \quad k_3 \in \mathbb{R} \quad t \in \mathbb{R} \quad \mathrm{xa} \in \mathbb{R} \quad \mathrm{xb} \in \mathbb{R} \quad 0 \le t\right)$

```
% the system matrices
matA = sym([0  0  1;
            0 -2 -1;
            4  0  2])
```

matA =

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & -2 & -1 \\ 4 & 0 & 2 \end{pmatrix}$$

```
matB = sym([1; 1; 2])
```

matB =

$$\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

The actual eigenvalues are

```
lambdaA = eig(matA)
```

lambdaA =

$$\begin{pmatrix} -2 \\ 1-\sqrt{5} \\ \sqrt{5}+1 \end{pmatrix}$$

```
% Note: the eigenvalues are real values
checkUNSTABILITY = any(lambdaA >0);

if checkUNSTABILITY
    disp('The system is UNSTABLE!')
end
```

```
The system is UNSTABLE!
```

The full state feedback matrix is

```
Kk = [k1, k2, k3]
```

Kk = $\begin{pmatrix} k_1 & k_2 & k_3 \end{pmatrix}$

Now the expression of the matrix $A+B\,K$ is simply

```
matAK = matA+matB*Kk
```

matAK =

$$\begin{pmatrix} k_1 & k_2 & k_3+1 \\ k_1 & k_2-2 & k_3-1 \\ 2\,k_1+4 & 2\,k_2 & 2\,k_3+2 \end{pmatrix}$$

The new characteristic polynomial can be determined using the MATLAB command **charpoly( )**

```
pAK_x = charpoly(matAK, x)
```

pAK_x = $x^3+(-k_1-k_2-2\,k_3)\,x^2+(4\,k_2-2\,k_1-8\,k_3-8)\,x+8\,k_2-8\,k_3-8$

Note that this polynomial depend also on the unknown coefficients **k1**, **k2** and **k3**.

The desired characteristic polynomial is

```
pD_x = expand((x+10)*(x+20)*(x+30))
```

pD_x = $x^3+60\,x^2+1100\,x+6000$

Let's extract the coefficients from the polynomials **pAK_x** and **pD_x**:

```
coeffsAK = coeffs(pAK_x, x)
```

coeffsAK = $\begin{pmatrix} 8\,k_2-8\,k_3-8 & 4\,k_2-2\,k_1-8\,k_3-8 & -k_1-k_2-2\,k_3 & 1 \end{pmatrix}$

```
coeffsD = coeffs(pD_x, x)
```

coeffsD = $\begin{pmatrix} 6000 & 1100 & 60 & 1 \end{pmatrix}$

The set of linear equations contains the following expressions

```
EQset = [coeffsAK(1) == coeffsD(1);
        coeffsAK(2) == coeffsD(2);
        coeffsAK(3) == coeffsD(3)]
```

EQset =

$$\begin{pmatrix} 8\,k_2 - 8\,k_3 - 8 = 6000 \\ 4\,k_2 - 2\,k_1 - 8\,k_3 - 8 = 1100 \\ -k_1 - k_2 - 2\,k_3 = 60 \end{pmatrix}$$

```
% Note: the coefficients of the highest degree monomials are equal and they
% do not depend on the parameters k1, k2, k3
```

Finally, the coefficients of the full state feedback are

```
kSOL = solve(EQset)
```

```
kSOL = struct with fields:
    k1: 4466
    k2: -1008
    k3: -1759
```